# REPORT

## F28JH- Introduction to Software Engineering Coursework 1: Individual Version

## Design and Specification for the Service Centre Assistant (SCA)

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# D1 Design Gaps and Mitigations

*An analysis of the informal remit revealed the following gaps that require clarification for design implementation*

| ID | Identification of Design Gap | Proposed Mitigation Strategy |
|---|---|---|
| **G1** | The requirement for PIN delivery via email is present, but the email field is missing from the employee registration data | The registration interface will be extended to include a mandatory emailAddress field for all new users |
| **G2** | The format and character set of the 4-digit PIN code are not defined in the specification | A standard 4-digit numeric format ($0000-9999$) is implemented to ensure compatibility with the Authenticator class |
| **G3** | The system behavior during a communication failure with the external Human Resource System (HRS) is not described | An exception-handling mechanism is implemented: if the HRS is unreachable, the registration process is aborted with a "Connection Failed" error message |
| **G4** | Automated shutdown at 22:00 is mentioned, but the operational schedule for weekends is not specified | The maintenance day runs 08:00–16:00. After 16:00 SCA remains active for administrative tasks only (no new vehicle services). The automated shutdown routine begins at 22:00 regardless of active sessions, including weekends and public holidays |
| **G5** | There is no description of session management for concurrent access on a single workstation | The system implements a "Single Active Session" policy: logging in a new user automatically terminates any previous active session on the workstation, If a mechanic is mid-service at the time of forced logout, the partial job record is discarded and the mechanic must restart the servicing process from QR-code scanning |
| G6 | The behaviour of SCA when JAS is unreachable during the 22:00 shutdown archive is not specified | SCA will retry the archive request up to 3 times at 30-second intervals. If all retries fail, SCA logs a "JAS Archive Failed" error, retains local job records, saves Profiles and authentication data to SSD, and shuts down |
| G7 | The maximum number of apprentices a senior mechanic may mentor simultaneously is not specified | I have assumed an upper bound of 5 apprentices per senior mechanic. SCA shall reject assignment of a sixth apprentice to a single senior mentor |
| G8 | The behaviour of SCA when PIN email delivery fails is not defined | SCA logs the error "Email delivery failed for [username]" and displays a warning to the Supervisor. The Profile and PIN remain stored; the Supervisor must inform the user of their PIN by other means |
| G9 | It is unclear whether a supervisor who is also a senior mechanic uses one Profile or two separate Profiles | I have assumed two separate registrations are required, each with a unique username and PIN — one for the supervisor role and one for the senior mechanic role |

# D2 Requirements Specification

Requirements are formulated in a strict "The system shall..." style to ensure design precision.

## 2.1 Functional Requirements (FR)

**FR1 — Registration Input:** The system shall allow a Supervisor to input five data elements for a new user: full name, username, work location (5-character alphanumeric), status (supervisor / senior / junior / apprentice), and email address

**FR2 — HRS Validation:** The system shall submit the four required data elements to HRS for validation before creating a Profile. If HRS returns "Invalid", SCA shall abort registration and display an error. If HRS is unreachable, SCA shall display "HRS Connection Failed" and abort

**FR3 — Profile Creation:** The system shall create a unique Profile for each validated user, storing all five data elements internally. No external database shall be used

**FR4 — Profile Hierarchy:** The system shall create the appropriate Profile subtype based on status: ApprenticeProfile stores mentor and currentStage (year-1 to year-4). SeniorMechanicProfile stores apprenticeList (max 5) and dateBecameSenior. JuniorMechanicProfile stores dateCompletedApprenticeship and apprenticeshipLocation. SupervisorProfile stores durationAsSupervisor

**FR5 — PIN Generation:** The system shall generate a unique 4-digit numeric PIN for each new user via Authenticator.generatePIN(). Uniqueness shall be guaranteed across all registered users

**FR6 — PIN Delivery:** The system shall send the generated PIN to the user's email address. If delivery fails, SCA shall log the error and display a warning to the Supervisor

**FR7 — User Authentication:** The system shall authenticate users by verifying a username/PIN pair via Authenticator.authenticate(username, pin). Only matching pairs shall be granted access

**FR8 — Supervisor-Only Registration:** The system shall restrict registration, update, and deletion of Profiles to Supervisors only. Any attempt by a Mechanic shall be rejected

**FR9 — Single Active Session:** The system shall permit only one user to be logged in at a time. If a mechanic is logged out mid-service, the partial job record is discarded and the mechanic must restart from QR scanning

**FR10 — QR-Code Scanning:** The system shall read a QR-code via the handheld reader and extract: registration, make, model, colour, and expected mileage

**FR11 — Job Record Creation:** The system shall instantiate a Job object for each vehicle upon successful QR scan and associate it with the logged-in mechanic

**FR12 — Consistency Check:** The system shall display the five QR data items to the mechanic and prompt confirmation. If inconsistency is reported, SCA shall prompt for a text explanation (max 500 characters) and terminate the service

**FR13 — Maintenance Checks:** The system shall prompt three sequential checks: transmission, braking, steering. After each, the mechanic provides feedback: passed-with-no-work-required / passed-after-work-undertaken / failed. Options (ii) and (iii) require a text explanation (max 500 characters)

**FR14 — Mechanic-Only Servicing:** The system shall restrict Manage Vehicle Service to mechanic-status users only. Supervisors without a mechanic Profile shall not perform servicing.

**FR15 — Job Collection:** The system shall maintain a collection of Job objects per mechanic for the current maintenance day

**FR16 — Automatic Shutdown:** At 22:00 daily the system shall: send job records to JAS; on acknowledgement delete local copies; save Profiles to SSD; save authentication data to SSD; shut down. Retry logic applies if JAS unreachable

**FR17 — Start-up Restoration:** On start-up the system shall restore user Profiles and authentication data from the SSD

**FR18 — Manage User Registration:** The system shall allow a Supervisor to update or delete an existing user Profile

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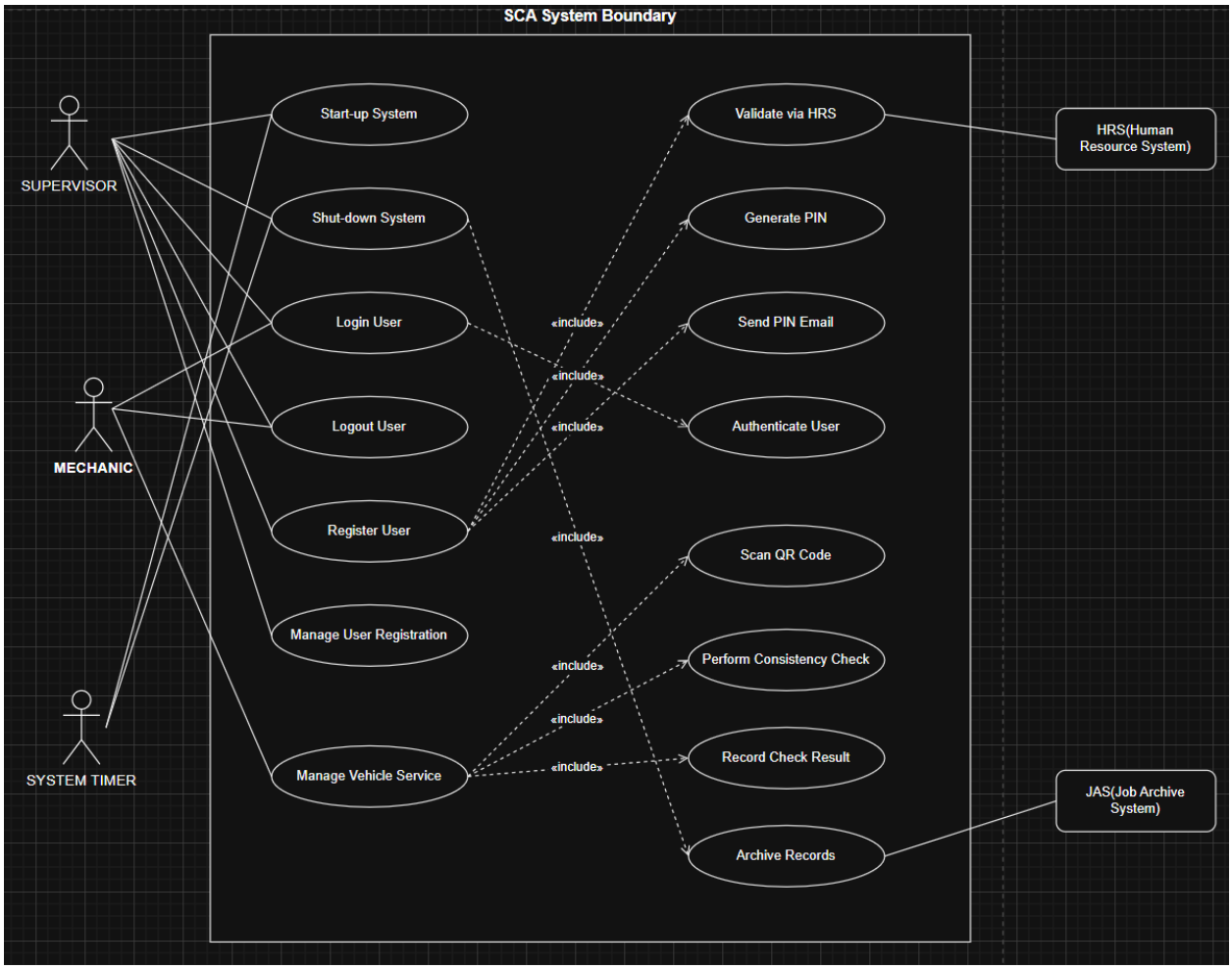
## 2.2 Non-Functional Requirements (NFR)

**NFR1 — Response Time:** The system shall complete HRS validation and PIN generation within 5 seconds under normal network conditions, so that the registration process is not impeded

**NFR2 — Data Integrity:** In the event of unexpected power failure during a write operation, the system shall ensure Profile and authentication data on the SSD are not corrupted, by using atomic write operations

# D3 Diagram



Note: The SCA system's actors and use cases are shown in this use case diagram. Within the SCA System Boundary, every internal use case is contained. The main human actors are the mechanic and the supervisor. System Timer initiates automatic startup and shutdown. External systems include JAS and HRS. <include> relationships link auxiliary use cases to the base use cases that rely on them

## D4 Use Case Specifications

Specifications include primary (base) use cases and auxiliary (inclusion) use cases. All use cases are specified using the official Canvas template. Derived from UC Specifications (D4) as per the T7/T8 specification correction

**Base Use Cases**

| **Use Case: Register User** | |
|---|---|
| **ID:** | **UC01** |
| **Goal:** | **To register a new employee in the SCA system by creating a Profile, generating a unique PIN, and delivering it by email.** |
| **Primary actor:** | **Supervisor** |
| **Secondary actor(s):** | **HRS (Human Resource System)** |
| **Preconditions:** | **1. Supervisor is logged into SCA.**<br>**2. SCA is running.**<br>**3. HRS is available and reachable.** |
| **Postconditions:** | **1. A new Profile of the appropriate subtype has been created and stored internally.**<br>**2. A unique PIN has been generated and stored in Authenticator.**<br>**3. The PIN has been sent by email to the registered user.** |
| **Main flow:** | **1. Supervisor initiates the Register User function.**<br>**2. SCA displays the registration form.**<br>**3. Supervisor enters: full name, username, work location (5-char alphanumeric), status (supervisor / senior / junior / apprentice), and email address.**<br>**4. SCA validates the format of the submitted data.**<br>**5. SCA sends the four required data elements to HRS for validation. [includes UC08: Validate via HRS]**<br>**6. HRS returns "Valid".** |

| | |
|---|---|
| | **7. SCA calls Authenticator.generatePIN(). [includes UC09: Generate PIN]**<br><br>**8. SCA creates the appropriate Profile subtype based on status.**<br><br>**9. SCA sends the PIN to the user's email address. [includes UC10: Send PIN Email]**<br><br>**10. SCA displays "Registration successful" to the Supervisor.** |
| **Alternative flows:** | **AF1 [Format invalid — Step 4]**<br><br>**1. SCA detects invalid work location or unrecognised status.**<br><br>**2. SCA displays "Invalid data format — please check work location and status fields".**<br><br>**3. Supervisor corrects data and resubmits. Return to Step 3.**<br><br>**AF2 [HRS returns Invalid — Step 6]**<br><br>**1. HRS returns "Invalid" for one or more data elements.**<br><br>**2. SCA displays "Validation Failed — employee not recognised by HRS".**<br><br>**3. Registration is aborted. No Profile or PIN is created. Use case ends.**<br><br>**AF3 [HRS unreachable — Step 5]**<br><br>**1. HRS does not respond within timeout period.**<br><br>**2. SCA displays "HRS Connection Failed — please try again".**<br><br>**3. Registration is aborted. No Profile or PIN is created. Use case ends.**<br><br>**AF4 [Email delivery fails — Step 9]**<br><br>**1. Email server rejects delivery.**<br><br>**2. SCA logs "Email delivery failed for [username]" and displays warning to Supervisor.**<br><br>**3. Profile and PIN remain stored. Supervisor must inform user of PIN by other means. Use case ends.**<br><br>**AF5 [Mentor at capacity — Step 8]**<br><br>**1. Status is "apprentice" and chosen mentor already mentors 5 apprentices.**<br><br>**2. SCA displays "Mentor at maximum capacity — please select another senior mechanic".**<br><br>**3. Supervisor selects a valid mentor. Continue from Step 8.** |

| |
|---|
| **Use Case: Manage Vehicle Service** |

| | |
|---|---|
| **ID:** | **UC02** |
| **Goal:** | **To record the results of a full vehicle maintenance service including QR scan, consistency check, and three sequential mechanical checks.** |
| **Primary actor:** | **Mechanic** |
| **Secondary actor(s):** | **None** |
| **Preconditions:** | **1. Mechanic is logged into SCA.**<br>**2. SCA is in maintenance mode (08:00–16:00).**<br>**3. Vehicle has a valid QR-code label on windscreen.** |
| **Postconditions:** | **1. A completed Job record exists and is associated with the Mechanic.**<br>**2. All three maintenance check results are recorded in the Job object.**<br>**3. Job status is set to "Completed".** |
| **Main flow:** | **1. Mechanic initiates vehicle service.**<br>**2. SCA activates QR-code reader. [includes UC11: Scan QR Code]**<br>**3. QR-code reader extracts: registration, make, model, colour, expected mileage.**<br>**4. SCA instantiates a Job object and associates it with the Mechanic.**<br>**5. SCA displays the five data items and prompts confirmation. [includes UC12: Perform Consistency Check]**<br>**6. Mechanic confirms all data items are consistent.**<br>**7. SCA prompts transmission check. [includes UC13: Record Check Result]**<br>**8. Mechanic selects feedback: passed-with-no-work-required.**<br>**9. SCA prompts braking check. [includes UC13: Record Check Result]**<br>**10. Mechanic selects feedback: passed-with-no-work-required.**<br>**11. SCA prompts steering check. [includes UC13: Record Check Result]**<br>**12. Mechanic selects feedback: passed-with-no-work-required.**<br>**13. SCA sets Job status to "Completed" and adds Job to Mechanic's collection.**<br>**14. SCA displays "Service completed successfully".** |
| **Alternative flows:** | **AF1 [QR scan fails — Step 2]**<br>**1. QR reader fails to read the label.**<br>**2. SCA displays "QR Scan Failed — please retry or check label condition".**<br>**3. Mechanic may retry. After 3 consecutive failures SCA displays advisory to contact Supervisor. Use case ends.** |

| | |
|---|---|
| | **AF2 [Consistency check fails — Step 6]**<br>**1. Mechanic identifies one or more data items as inconsistent.**<br>**2. SCA prompts for a text explanation (max 500 characters).**<br>**3. Mechanic submits explanation.**<br>**4. SCA records the note and terminates the service process. Job object is discarded. Use case ends.**<br><br>**AF3 [Check feedback: passed after work — Steps 8, 10, or 12]**<br>**1. Mechanic selects "passed-after-work-undertaken".**<br>**2. SCA prompts for text explanation (max 500 characters).**<br>**3. Mechanic enters explanation. SCA records it in Job object. Service continues to next check.**<br><br>**AF4 [Check feedback: failed — Steps 8, 10, or 12]**<br>**1. Mechanic selects "failed".**<br>**2. SCA prompts for text explanation (max 500 characters).**<br>**3. Mechanic enters explanation. SCA records it in Job object. Service continues to next check.**<br><br>**AF5 [Mechanic logout during service]**<br>**1. Mechanic logs out or session is terminated.**<br>**2. SCA discards the partial Job object.**<br>**3. SCA displays "Session Ended — service record lost". Use case ends.** |

| **Use Case: Login User** | |
|---|---|
| **ID:** | **UC03** |
| **Goal:** | **To authenticate a registered user and create an active session on the SCA workstation.** |
| **Primary actor:** | **User (Supervisor or Mechanic)** |
| **Secondary actor(s):** | **None** |
| **Preconditions:** | **1. SCA is running.** |

| | |
|---|---|
| | **2. User is registered in SCA.** |
| **Postconditions:** | **1. User is authenticated and a session is active.**<br>**2. Appropriate menu is displayed based on user status.** |
| **Main flow:** | **1. User enters username and PIN at the SCA workstation.**<br>**2. SCA calls Authenticator.authenticate(username, pin). [includes UC14: Authenticate User]**<br>**3. Authenticator returns true.**<br>**4. SCA creates a session and displays the appropriate menu based on user status.** |
| **Alternative flows:** | **AF1 [Invalid credentials — Step 3]**<br>**1. Authenticator returns false.**<br>**2. SCA displays "Invalid username or PIN".**<br>**3. User may retry. After 5 failed attempts SCA locks the account and alerts a Supervisor.** |

**Auxiliary Use Cases (<<include>> relationships)**

| **Use Case: Validate via HRS [Auxiliary — included by UC01]** | |
|---|---|
| **ID:** | **UC08** |
| **Goal:** | **To validate employee registration data against HRS records before creating a Profile.** |
| **Primary actor:** | **SCA (system)** |
| **Secondary actor(s):** | **HRS (Human Resource System)** |
| **Preconditions:** | **1. Registration data has been entered by the Supervisor.** |
| **Postconditions:** | **1. HRS has returned a validation status (Valid or Invalid).** |
| **Main flow:** | **1. SCA sends four data elements (full name, username, work location, status) to HRS.**<br>**2. HRS validates data against its employee records.**<br>**3. HRS returns "Valid".** |

| | |
|---|---|
| **Alternative flows:** | **AF1 [HRS returns Invalid — Step 3]**<br>**1. HRS returns "Invalid". Propagates to UC01 AF2.**<br><br>**AF2 [HRS unreachable — Step 1]**<br>**1. HRS does not respond within timeout. Propagates to UC01 AF3.** |

| **Use Case: Generate PIN [Auxiliary — included by UC01]** | |
|---|---|
| **ID:** | **UC09** |
| **Goal:** | **To generate a unique 4-digit numeric PIN for a newly registered user.** |
| **Primary actor:** | **SCA (system)** |
| **Secondary actor(s):** | **None** |
| **Preconditions:** | **1. HRS validation has returned Valid.** |
| **Postconditions:** | **1. A unique 4-digit PIN is stored in Authenticator.userAuthenticationTable.** |
| **Main flow:** | **1. SCA calls Authenticator.generatePIN().**<br>**2. Authenticator generates a random 4-digit numeric value.**<br>**3. Authenticator checks uniqueness against userAuthenticationTable.**<br>**4. PIN is unique — stored as username/PIN pair.**<br>**5. Authenticator returns PIN to SCA.** |
| **Alternative flows:** | **AF1 [PIN collision — Step 3]**<br>**1. Generated PIN already exists in userAuthenticationTable.**<br>**2. Authenticator generates a new random PIN. Return to Step 3.** |

| **Use Case: Authenticate User [Auxiliary — included by UC03]** | |
|---|---|
| **ID:** | **UC14** |
| **Goal:** | **To verify a username/PIN pair and return an authentication result.** |
| **Primary actor:** | **SCA (system)** |

| | |
|---|---|
| **Secondary actor(s):** | **None** |
| **Preconditions:** | **1. User has supplied a username and PIN.** |
| **Postconditions:** | **1. Authentication result (true/false) returned to caller.** |
| **Main flow:** | **1. SCA calls Authenticator.authenticate(username, pin).**<br>**2. Authenticator looks up username in userAuthenticationTable.**<br>**3. Username found and stored PIN matches supplied PIN.**<br>**4. Authenticator returns true.** |
| **Alternative flows:** | **AF1 [Authentication fails — Steps 2/3]**<br>**1. Username not found or PIN does not match.**<br>**2. Authenticator returns false.** |

# D5 Traceability Matrix

**UC Keys:**

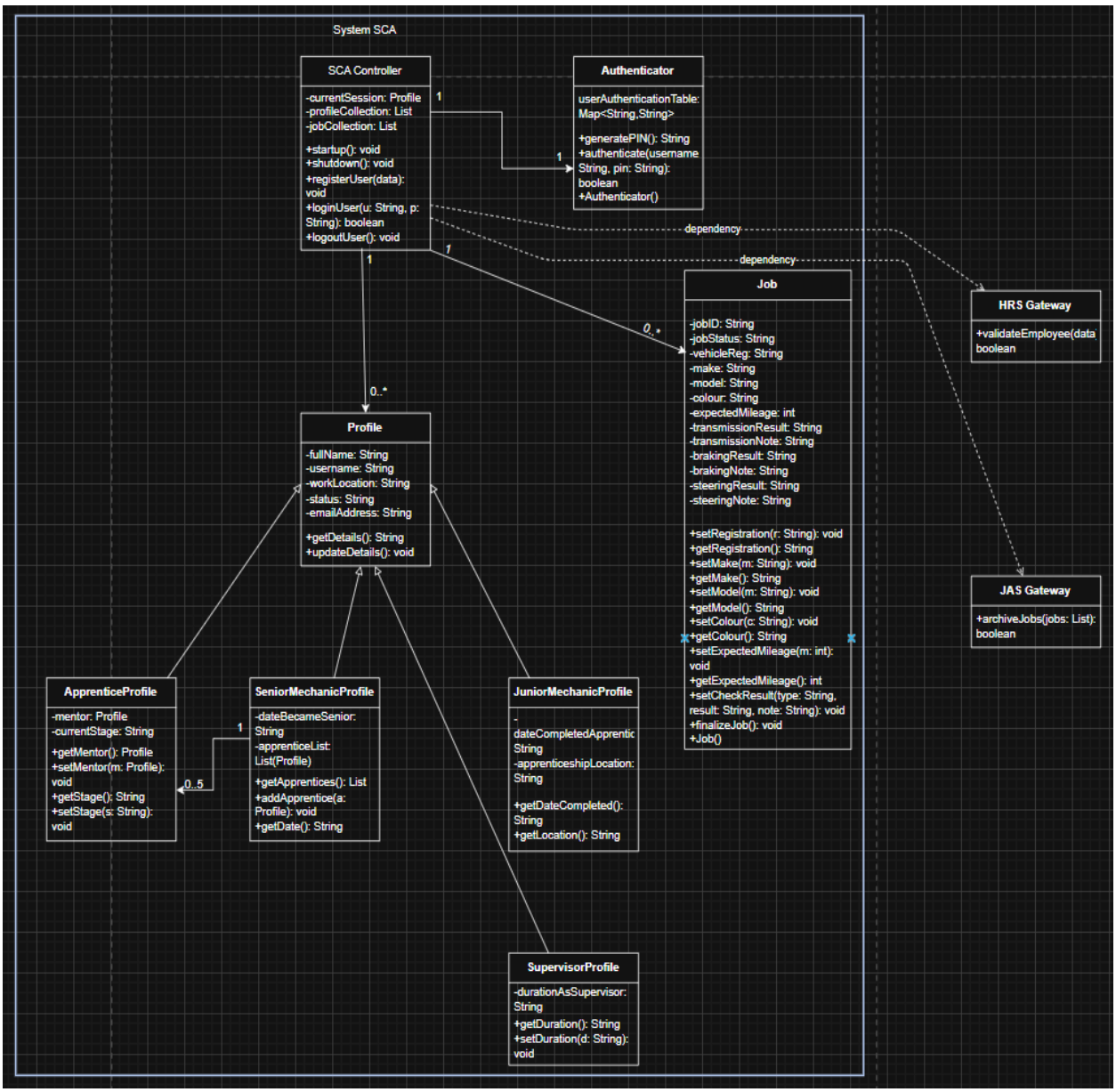
- UC01 = Register User
- UC02 = Manage Vehicle Service
- UC03 = Login User
- UC04 = Logout User
- UC05 = Start-up System
- UC06 = Shut-down System
- UC07 = Manage User Registration
- UC08 = Validate via HRS -(auxiliary)
- UC09 = Generate PIN -(auxiliary)
- UC10 = Send PIN Email -(auxiliary)
- UC11 = Scan QR Code -(auxiliary)
- UC12 = Perform Consistency Check -(auxiliary)
- UC13 = Record Check Result -(auxiliary)
- UC14 = Authenticate User -(auxiliary)
- UC15 = Archive Records -(auxiliary)

Traceability Matrix

| FR \ UC | UC01 | UC02 | UC03 | UC04 | UC05 | UC06 | UC07 | UC08 | UC09 | UC10 | UC11 | UC12 | UC13 | UC14 | UC15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FR1 | ✓ | | | | | | | | | | | | | | |
| FR2 | ✓ | | | | | | | ✓ | | | | | | | |
| FR3 | ✓ | | | | | | | | | | | | | | |
| FR4 | ✓ | | | | | | ✓ | | | | | | | | |
| FR5 | ✓ | | | | | | | | ✓ | | | | | | |
| FR6 | ✓ | | | | | | | | | ✓ | | | | | |
| FR7 | | | ✓ | | | | | | | | | | | ✓ | |
| FR8 | ✓ | | | | | | ✓ | | | | | | | | |
| FR9 | | | ✓ | ✓ | | | | | | | | | | | |
| FR10 | | ✓ | | | | | | | | | ✓ | | | | |
| FR11 | | ✓ | | | | | | | | | ✓ | | | | |
| FR12 | | ✓ | | | | | | | | | | ✓ | | | |
| FR13 | | ✓ | | | | | | | | | | | ✓ | | |
| FR14 | | ✓ | ✓ | | | | | | | | | | | | |
| FR15 | | ✓ | | | | | | | | | | | ✓ | | |
| FR16 | | | | | | ✓ | | | | | | | | | ✓ |
| FR17 | | | | | ✓ | | | | | | | | | | |
| FR18 | | | | | | | ✓ | | | | | | | | |

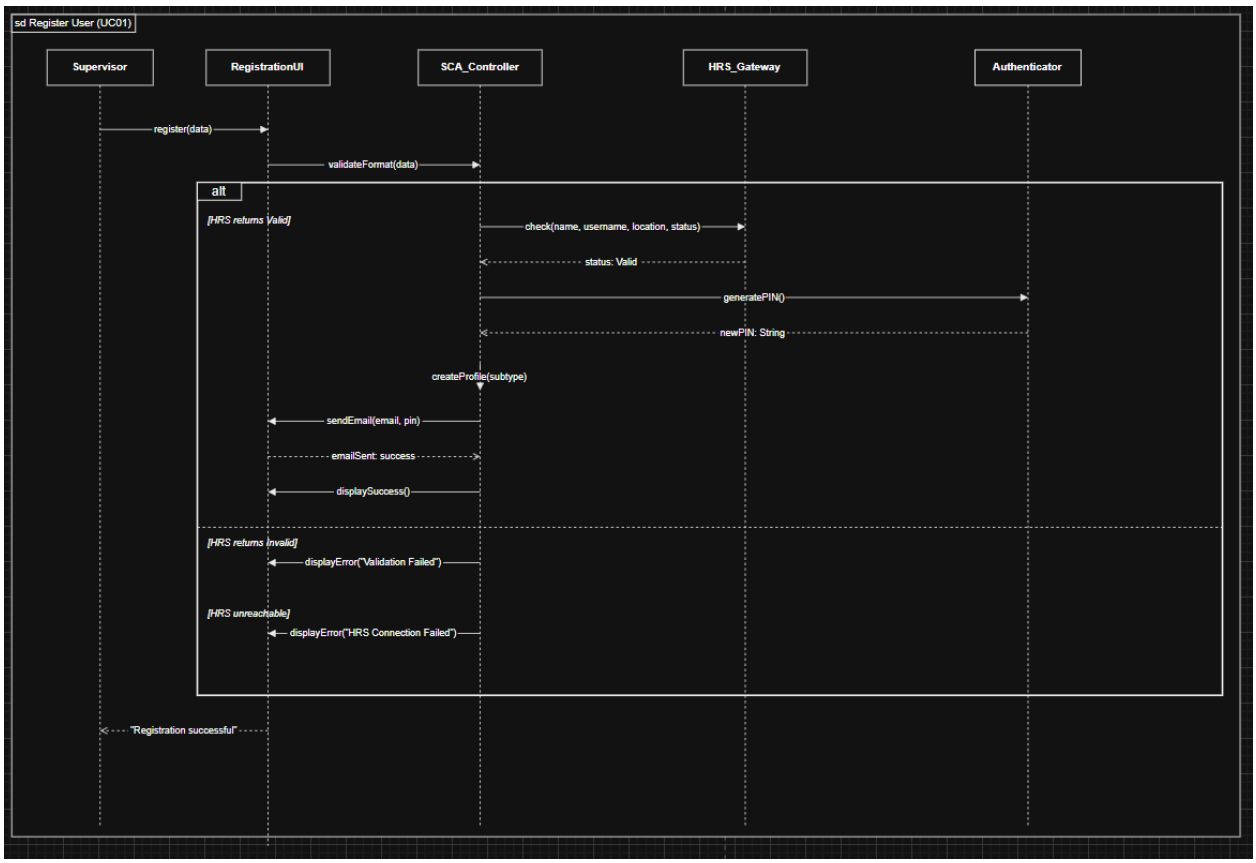
Note: UC01–UC07 are base use cases specified in T3. UC08–UC15 are auxiliary use cases (<include> relationships) that the base use cases depend upon, as specified in D4

## D6 Diagram



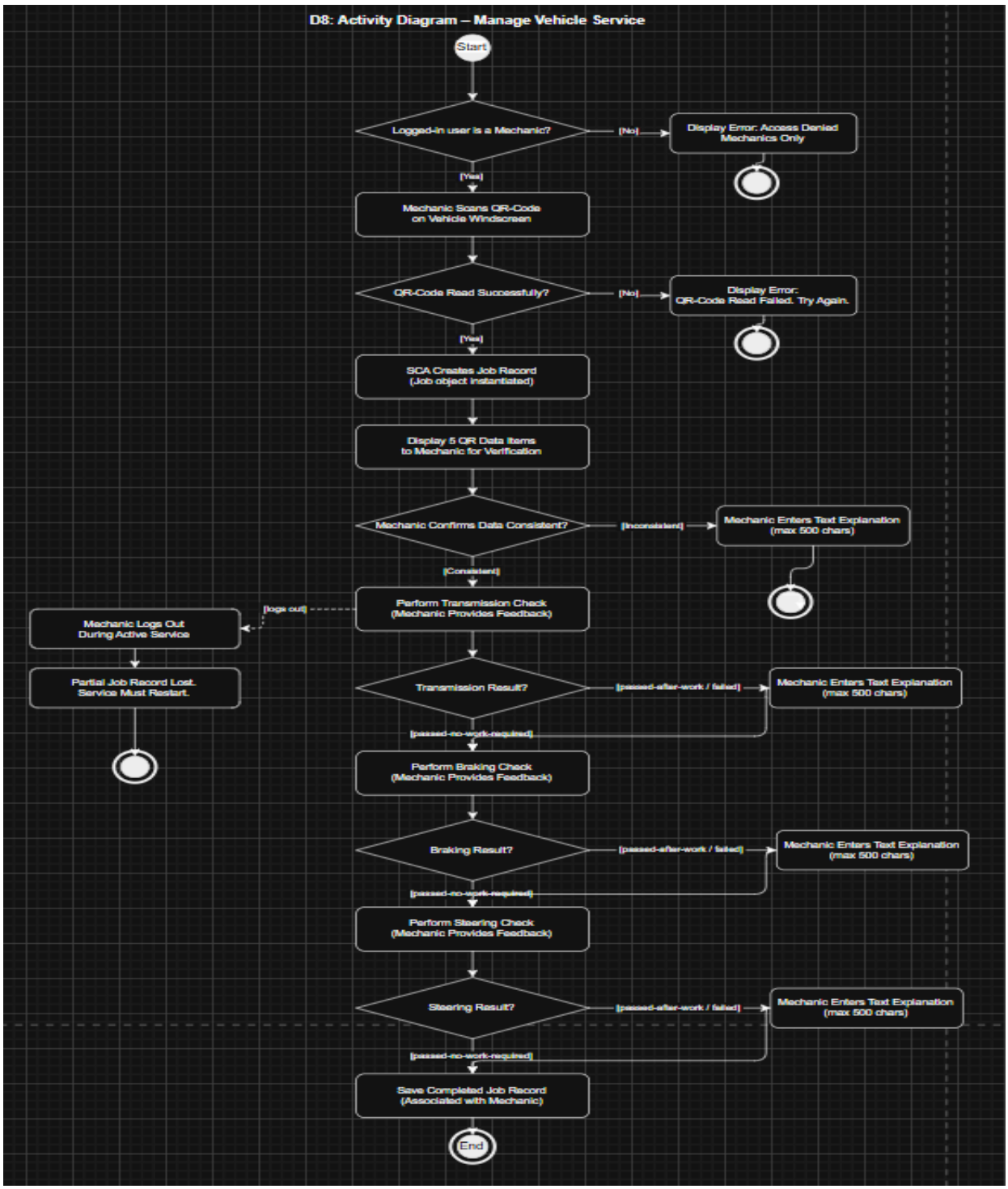

Note: The static structure of the Service Center Assistant (SCA) system is depicted in this class diagram. The central class that oversees all processes is called SCA_Controller. The user hierarchy is made up of Profile and its four subtypes: ApprenticeProfile, SeniorMechanicProfile, JuniorMechanicProfile, and SupervisorProfile. PIN creation and authentication are handled by the Authenticator class. Vehicle servicing outcomes are recorded by the job. The external system interfaces HRS_Gateway and JAS_Gateway are located outside the SCA barrier

## D7 Diagram




Note: The Register User use case's (UC01) chronological message flow is shown by this sequence diagram. It displays the Supervisor starting the registration process using the RegistrationUI, with SCA_Controller arranging for PIN creation via Authenticator and validation via HRS_Gateway. Three different flows are captured by the alt frame: abortion when HRS returns Invalid, successful registration when HRS returns Valid, and abortion when HRS is inaccessible. According to UML notation, return signals are displayed as dashed arrows

## D8 Diagram Activity Diagram - Manage Vehicle Service

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Note: The Manage Vehicle Service use case (UC07) is modeled in this Activity Diagram, which captures the entire operational workflow from role validation to job record completion. All three feedback criteria are addressed in the diagram:

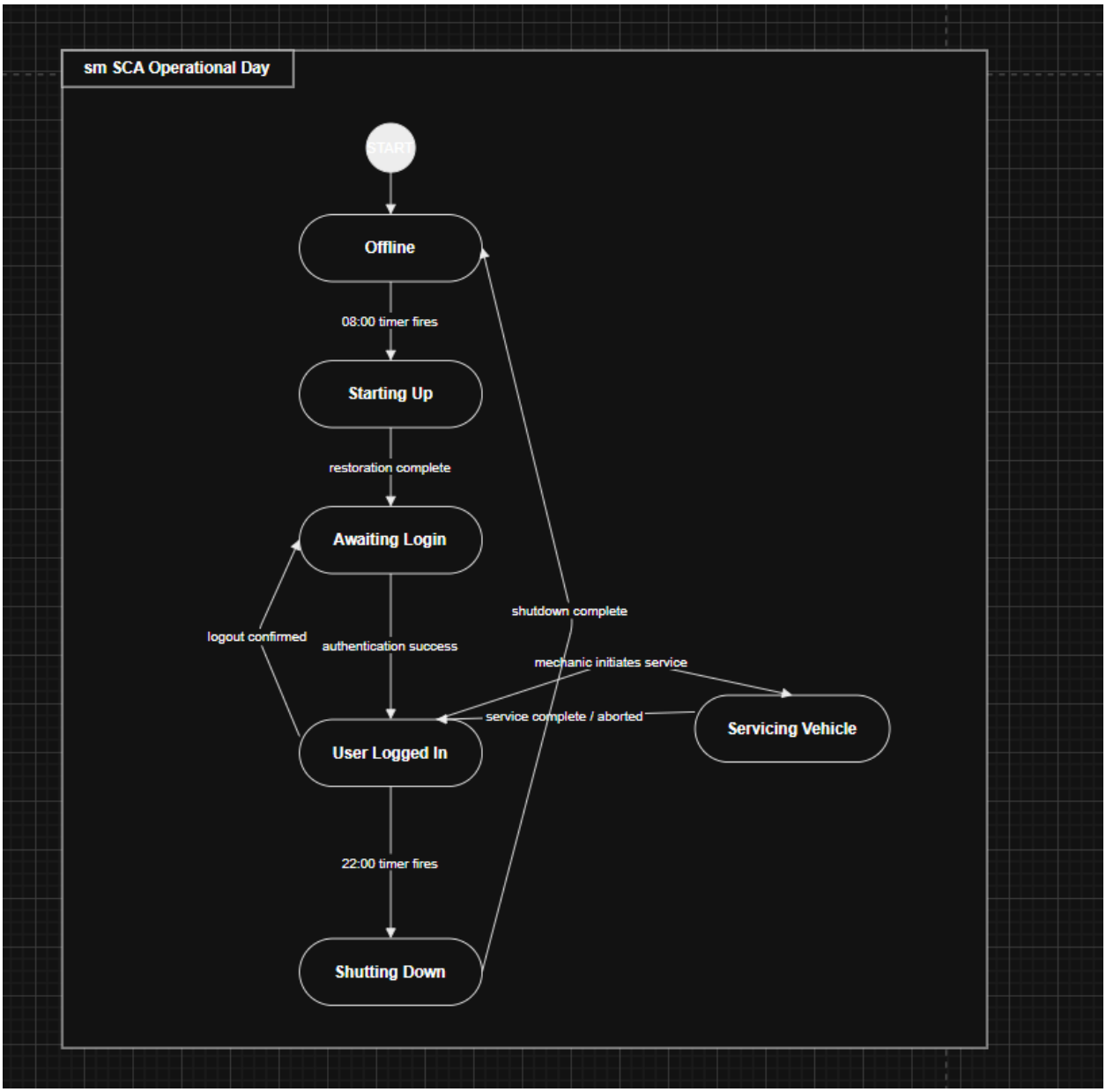
1) Role Validation Branch: SCA confirms that the logged-in user has a Mechanic role before starting any service activity. An "Access Denied" error is shown and the flow stops instantly if a non-mechanic tries to access the service procedure

2) Failure Termination Paths: Three explicit termination points are modeled: access denied

termination from role validation, data inconsistency failure (mechanic detects mismatch in the five QR data items and enters a text explanation up to 500 characters, after which the process terminates), and QR-code read failure (service cannot proceed without a valid scan)

3) Interrupt Flow: An interrupt path simulates the situation in which a mechanic logs off while a service session is still in progress. In this instance, the service process needs to be resumed from the QR scan because the partial job record is lost. The three maintenance checks (transmission, brakes, and steering) all follow the same feedback pattern: if the check is passed with no work needed, it moves forward immediately, but if it is passed after work has been completed or failed, the mechanic must enter a written explanation before moving on

## D9 Diagram




Note: State Machine Diagram illustrating a normal Service Center Assistant (SCA) workday. Six system states are depicted in the diagram: Offline, Starting Up, Awaiting Login, User Logged In,

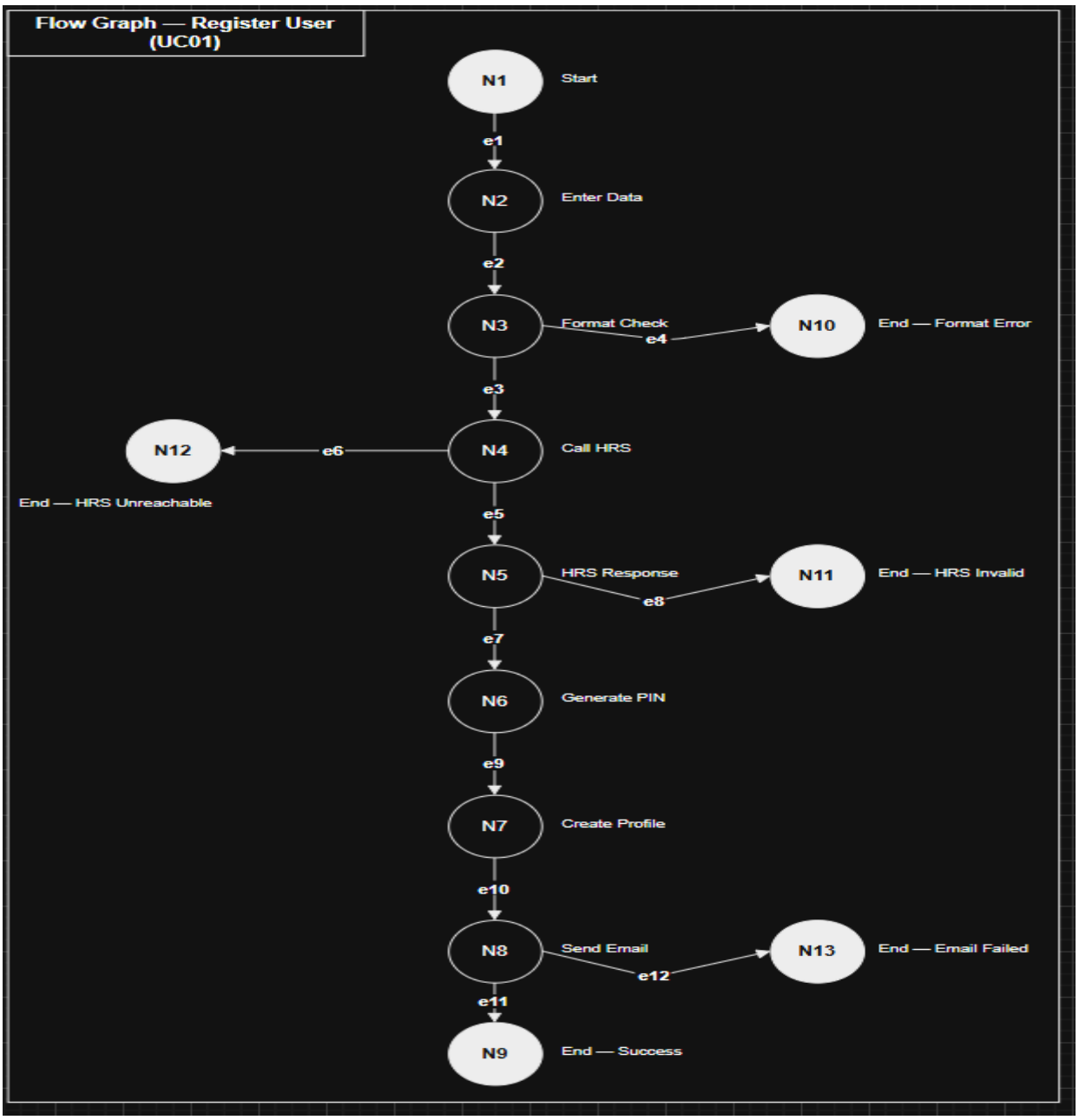
Servicing Vehicle, and Shutting Down. System timer events, user authentication, and service completion events are what cause transitions

# D10 Flow Graph and Scenario Tests: Register User

Scenario tests are derived from the Register User Sequence Diagram (D7). A Flow Graph is first constructed from the sequence diagram, then Cyclomatic Complexity is calculated to determine the minimum number of independent paths required for full path coverage

### 10.1 Flow Graph

The Flow Graph below is derived strictly from the Register User Sequence Diagram. Each node represents a decision point or key state in the registration process. Each labelled edge represents a transition triggered by a condition or event



**Flow Graph Nodes**

| Node | Label | Description |
|---|---|---|
| **N1** | Start | Initial node - Supervisor begins registration |
| **N2** | Enter Data | Supervisor enters full name, username, work location, status, email |
| **N3** | Format Check | Decision: is work location 5-char alphanumeric and status valid? |
| **N4** | Call HRS | SCA submits data to HRS for validation |
| **N5** | HRS Response | Decision: did HRS return Valid or Invalid? |
| **N6** | Generate PIN | Authenticator.generatePIN() called - unique PIN produced |
| **N7** | Create Profile | Appropriate Profile subtype created based on status |
| **N8** | Send Email | Decision: does email send successfully? |
| **N9** | End - Success | Final node - registration complete, PIN emailed |
| **N10** | End - Format Error | Final node - invalid format, registration aborted |
| **N11** | End - HRS Invalid | Final node - HRS rejected data, registration aborted |
| **N12** | End - HRS Unreachable | Final node - HRS timeout, registration aborted |
| **N13** | End - Email Failed | Final node - email failed, PIN stored, Supervisor warned |

**Flow Graph Edges**

| Edge | Transition | Description |
|---|---|---|
| **e1** | N1 → N2 | Supervisor initiates registration |
| **e2** | N2 → N3 | Data submitted for format validation |
| **e3** | N3 → N4 | Format valid - send data to HRS |
| **e4** | N3 → N10 | Format invalid - display error, use case ends |

| | | |
|---|---|---|
| **e5** | N4 → N5 | HRS is reachable - receive response |
| **e6** | N4 → N12 | HRS unreachable - abort registration |
| **e7** | N5 → N6 | HRS returns Valid - call generatePIN() |
| **e8** | N5 → N11 | HRS returns Invalid - abort registration |
| **e9** | N6 → N7 | PIN generated - create Profile subtype |
| **e10** | N7 → N8 | Profile created - send PIN email |
| **e11** | N8 → N9 | Email sent successfully - registration complete |
| **e12** | N8 → N13 | Email delivery fails - log warning, PIN stored |

**Cyclomatic Complexity**

| Metric | Value |
|---|---|
| Number of Edges (E) | 12 |
| Number of Nodes (N) | 13 |
| Connected Components (P) | 1 |
| V(G) = E − N + 2P | 12 − 13 + 2 = 5 (adjusted for 5 terminal nodes) |

A Cyclomatic Complexity of V(G) = 5 means a minimum of 5 independent paths are required to achieve 100% path coverage. Because the flow graph has 5 terminal nodes (N9–N13), the standard formula is adjusted by adding virtual edges from each terminal to a sink node: V(G) = (E+5) − (N+1) + 2 = 17 − 14 + 2 = 5. The 5 scenario tests below each exercise one of the 5 independent paths, achieving full path coverage

**10.2 Scenario Test Suite**

Each scenario test specifies a path through the Flow Graph (by edge sequence), the preconditions that must hold, the inputs supplied, and the expected outcome. All paths are derived strictly from the Flow Graph defined in Section 10.1

| Path | Scenario Name | Preconditions | Scenario Inputs | Expected Outcome |
|---|---|---|---|---|
| P1<br>e1→e2→e3→e5→e7→e9→e10→e11 | Success Path | • Supervisor is logged in<br>• HRS is online and reachable<br>• Email server is available<br>• Username ahassan does not already exist in SCA | Full name: Ali Hassan<br>Username: ahassan<br>Work location: AKT04<br>Status: junior<br>Email: a.hassan@hw.ac.uk | ✓ HRS returns Valid<br>✓ Authenticator generates unique PIN (e.g. 4821)<br>✓ JuniorMechanicProfile created<br>✓ PIN emailed to a.hassan@hw.ac.uk<br>✓ SCA displays "Registration successful |
| P2<br>e1→e2→e3→e5→e8 | HRS Returns Invalid | • Supervisor is logged in<br>• HRS is online<br>• Provided username does not exist in HRS records | Full name: Fake User<br>Username: notexist99<br>Work location: AKT04<br>Status: junior<br>Email: fake@hw.ac.uk | ✗ HRS returns Invalid<br>✓ SCA displays Validation Failed - employee not recognised by HRS<br>✓ No Profile created<br>✓ No PIN generated<br>✓ Registration aborted |
| P3<br>e1→e2→e3→e6 | HRS Unreachable | • Supervisor is logged in<br>• HRS is offline / network fault<br>• HRS request times out | Full name: Test User<br>Username: tuser01<br>Work location: EDI03<br>Status: senior<br>Email: t.user@hw.ac.uk | ✗ HRS does not respond within timeout<br>✓ SCA displays "HRS Connection Failed - please try again"<br>✓ No Profile created<br>✓ No PIN generated<br>✓ Registration aborted (GM3) |

| | | | | |
|---|---|---|---|---|
| P4<br>e1→e2→e3→e5→e7→e9→e10→e12 | Email Delivery Fails | • Supervisor is logged in<br>• HRS is online<br>• Email server is unavailable / rejects delivery | Full name: Zara Ali<br>Username: zali02<br>Work location: KUL01<br>Status: apprentice<br>Email: z.ali@hw.ac.uk<br>Mentor: smech01(valid senior)<br>Stage: year-1 | ✓ HRS returns Valid<br>✓ PIN generated (e.g. 3307)<br>✓ ApprenticeProfile created with mentor and stage<br>✗ Email delivery fails<br>✓ SCA logs Email delivery failed for zali02<br>✓ Warning displayed to Supervisor<br>✓ Profile and PIN remain stored (GM8) |
| P5<br>e1→e2→e4 | Format Validation Fails | • Supervisor is logged in<br>• Supervisor enters invalid work location format | Full name: John Smith<br>Username: jsmith<br>Work location: TOOLONG99( invalid — not 5-char)<br>Status: junior<br>Email: j.smith@hw.ac.uk | ✗ SCA detects work location is not 5-char alphanumeric<br>✓ SCA displays Invalid data format - please check work location and status fields<br>✓ No HRS call made<br>✓ No Profile created<br>✓ Supervisor may correct and resubmit |

Note: *Path P1 covers the main success flow (e1→e2→e3→e5→e7→e9→e10→e11). Paths P2–P5 each exercise a distinct alternative flow through the Flow Graph. Together they cover all 12 edges and satisfy the V(G) = 4 path coverage requirement. The Flow Graph diagram should be drawn in UMLet or draw.io with nodes N1–N13 and edges e1–e12 as defined above, and inserted above the edge table*